# Hands-on Examination 1 (contributed by J. B. Tuazon)

## Pikachu's Pokedex Entry

Professor Oak's Pokemon database got corrupted, and it is up to you to re-enter the Pokedex entries. You are tasked to input Pikachu's Pokedex number, weight in pounds, base statistics which includes its HP, Attack, Defense, and Speed. You are also required to show the statistics of the Pikachu given its projected level and its shiny variety.

### Tasks:

1. **Display Pikachu's details (10 points)**
   a. Implement a function called `displayPikachu()` which will show the following details:
      - Pokedex No: `<depends on the user input>`
        1. This number will be an input from `int main()` and will be passed on as an argument to `displayPikachu()`
      - Name: Pikachu
      - Type: Electric
      - Ability: Static
      - Weight in Kilograms: `<depends on the user input>`
        1. This value will be an input from `int main()` and will be passed on as an argument to `displayPikachu()`. This value will be entered in pounds but must be converted to kilograms within the function.
      - Description: This Pokemon has electricity-storing pouches on its cheeks.

   b. This function will be called within `int main()` and does not return a value.
2. **Display Pikachu's Base Stats (5 Points)**
   a. Implement a function called `displayBaseStats()` which will show the base stats of Pikachu. These stats include the following:
      - Base HP
      - Base Attack
      - Base Defense
      - Base Speed

   b. Ensure that your output is properly formatted. You may follow the sample screenshots or create your own. This function will be called within `int main()`.
   c. Return the total base stats.
   d. All data whose type is either `float` or `double` must be displayed with two decimal places.
3. **Compute and display Pikachu's levelled-up stats (15 points)**
   a. Implement a function called `displayLevelledUpStats()`. The levelled-up stats can be computed with the following considerations:
      - Every one level up increases the HP by `15%`

1. For example, if the base stat for HP is `100.00`, then every level up increases the HP by `15.00`. If the indicated level is `10`, then the additional HP is `150.00`, making the total HP `250.00` at level 10.

- Every one level up increases the Attack by `5%`
    1. For example, if the base stat for Attack is `100.00`, then every level up increases the Attack by `5.00`. If the indicated level is `10`, then the additional Attack is `50.00` making the total Attack `150.00` at level `10`.
- Every one level up increases the Defense by `10%`
    1. For example, if the base stat for Defense is `100.00`, then every level up increases the Defense by `10`. If the indicated level is `10`, then the additional Attack is `100.00`, making the total Defense `200.00` at level `10`.
- Every one level up increases the Speed by `25%`
    1. For example, if the base stat for Speed is `100.00`, then every level up increases the Attack by `25.00`. If the indicated level is `10`, then the additional Speed is `250.00`, making the total Speed `350.00` at level `10`.

b. Take note that the percentage increase is not compounded. This means that the base statistics will always be the basis of the stats increase.

c. This function will be called within `int main()`.

d. Return the total levelled-up stats.

e. All data whose type is either `float` or `double` must be displayed with two decimal places.

**4. Compute and display Pikachu's shiny variety's levelled-up stats (10 points)**

a. Implement a function called `displayShinyStats()`.

b. To compute for the shiny stats, add a percentage increase to the levelled-up stats by multiplying them by the multiplier bonus from `int main()`.

- For example, if the levelled-up stats for HP is `250.00` and the multiplier bonus is `10%`, the shiny stat for HP is `275.00`.

c. This function must be called by the `displayLevelledUpStats()` function.

d. All data whose type is either `float` or `double` must be displayed with two decimal places.

**5. Compute Pikachu's friendship bonus stats (10 Points)**

a. Implement a `function` called `computeFriendshipBonus();`

b. To compute for the bonus base stats:

- `15%` increase in the base HP
- `5%` increase in the base Attack
- `5%` increase in the base Defense
- `20%` increase in the base Speed

c. This function must be called by `int main()` and the arguments passed must be the memory addresses of the variables of the base stats. This means that the parameters of the function must be declared as pointers.

d. There are no displays in this function.
e. There is no return value for this function.

## Sample outputs:

```
Inputs:
Pikachu's Pokedex No: 25↵
Weight in pounds: 13.20↵
Base HP: 100.00↵
Base Attack: 100.00↵
Base Defense: 100.00↵
Base Speed: 100.00↵
Input the level of the Pokemon: 10↵
Input the multiplier bonus (in percent) of the shiny version: 10.00↵
```

```
Pikachu's Pokedex Number: 25
Weight in pounds: 13.2
------------------------------------------------------------------------------------------------------------
Input the base stats of Pikachu
Base HP: 100.00
Base Attack: 100.00
Base Defense: 100.00
Base Speed: 100.00
------------------------------------------------------------------------------------------------------------
Input the projected level of the Pokemon: 10
Input the multiplier bonus (in percent) of the shiny version: 10.00
------------------------------------------------------------------------------------------------------------
Pokedex No:     25
Name:           Pikachu
Type:           Electric
Ability:        Static
Weight:         6.00KGs
Description:    This Pokemon has electricity-storing pouches on its cheeks.
------------------------------------------------------------------------------------------------------------
                          HP             Attack          Defense          Speed            Total
------------------------------------------------------------------------------------------------------------
Base Stats               100.00          100.00          100.00          100.00           400.00
Level 10 Stats           250.00          150.00          200.00          350.00           950.00
Level 10 Shiny Stats     275.00          165.00          220.00          385.00          1045.00
------------------------------------------------------------------------------------------------------------
Average Base Stats:              100.00
Average Level 10 Stats:          237.50
Average Shiny Level 10 Stats:    261.25
------------------------------------------------------------------------------------------------------------
Friendship Bonus Base Stats:
Base HP:        115.00
Base Attack:    105.00
Base Defense:   105.00
Base Speed:     120.00
```

```
Inputs:
Pikachu's Pokedex No: 25↵
Weight in pounds: 13.20↵
Base HP: 35.00↵
Base Attack: 55.00↵
Base Defense: 40.00↵
Base Speed: 90.00↵
Input the level of the Pokemon: 25↵
Input the multiplier bonus (in percent) of the shiny version: 20.00↵
```

```
Pikachu's Pokedex Number: 25
Weight in pounds: 13.2
----------------------------------------------------------------------------------------------------------
Input the base stats of Pikachu
Base HP: 35.00
Base Attack: 55.00
Base Defense: 40.00
Base Speed: 90.00
----------------------------------------------------------------------------------------------------------
Input the projected level of the Pokemon: 25
Input the multiplier bonus (in percent) of the shiny version: 20.00
----------------------------------------------------------------------------------------------------------
Pokedex No:     25
Name:           Pikachu
Type:           Electric
Ability:        Static
Weight:         6.00KGs
Description:    This Pokemon has electricity-storing pouches on its cheeks.
----------------------------------------------------------------------------------------------------------
                          HP            Attack          Defense         Speed           Total
----------------------------------------------------------------------------------------------------------
Base Stats               35.00           55.00           40.00           90.00           220.00
Level 25 Stats          166.25          123.75          140.00          652.50          1082.50
Level 25 Shiny Stats    199.50          148.50          168.00          783.00          1299.00
----------------------------------------------------------------------------------------------------------
Average Base Stats:               55.00
Average Level 25 Stats:          270.63
Average Shiny Level 25 Stats:    324.75
----------------------------------------------------------------------------------------------------------
Friendship Bonus Base Stats:
Base HP:        40.25
Base Attack:    57.75
Base Defense:   42.00
Base Speed:     108.00
```

To debug and test your program using a user-defined module (.c) and with the main() in a separate file, you need to do the following:

1. You need to have both files stated above in the same folder (directory).
2. You can compile and run the source code using the IDE (DevC++) or run it in command prompt with the following command : "`gcc -Wall -std=c99 pikachu-main.c -o pikachu-main`".
3. Run the code by typing in "`pikachu-main`".

**DELIVERABLES:**

Your C program source file which contains the functions must be named "`pikachu-lastname.c`" with your own last name as filename. For example, if your lastname is SANTOS, then the source file should be named as "`pikachu-santos.c`". Upload your source files in AnimoSpace before the indicated submission time.

**TESTING & SCORING:**

- Your program will be compiled via `gcc -Wall -std=c99`. Thus, for each function that does not compile successfully, the score for that function is 0.
- Your program will be tested by your instructor with other sample inputs (different from the ones given to you) and with function calls of different parameter values. It is also possible for the instructor to test your "`pikachu-santos.c`" file (the functions) using a different `main()` program.
- Full credit will be given for the function only if the student's implementation is all correct for all the test values used by the instructor during checking AND only if the student's implementation complied with the requirement and did not violate restrictions. Deductions will be given if not all test cases produce correct results OR if restrictions were not followed.

ASSUMPTIONS:

- All base stats inputs are any positive real numbers starting from 1.00.
- The projected level input is positive starting from one (1).
- The multiplier bonus for the shiny variety is zero and above.